PES UNIVERSITY, BANGALORE

Department of Computer Science and Engineering

SOFTWARE TEST PLAN (STP)

Event Management System

Project	Event Management System
Version	1.0
Team	Team 7
Authors	Aryan Mittal, Ashmita Chaki, Ashwath Rohan, Atharv Kulkarni
SRNs	PES2UG24CS089, PES2UG24CS090, PES2UG24CS091, PES2UG24CS092
Date	16-09-2026
Status	Draft / Initial Test Plan

Revision History

Version	Date	Author	Change Summary	Approval
1.0	16-09-2026	Aryan Mittal, Ashmita Chaki, Ashwath Rohan, Atharv Kulkarni	Initial Test Plan for Event Management System based on SRS v1.0	Pending

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			
Project Team Member	Ashmita Chaki	Ashmita Chaki chakiashmita@gmail.com	16-09-2026
Project Team Member	Aryan Mittal	Aryan Mittal aryanmittal2107@gmail.com	16-09-2026
Project Team Member	Ashwath Rohan	Ashwath Rohan ashwathrohan16@gmail.com	16-09-2026
Project Team Member	Atharv Kulkarni	Atharv Kulkarni starcak06@gmail.com	16-09-2026

1. Introduction

Purpose: This test plan describes the testing objectives, scope, strategy, environment, test cases, traceability, risks, metrics, and responsibilities for Event Management System (EMS) v1.0.

Scope: Testing will cover account registration and authentication, role-based access, event creation/editing/publishing/cancellation, event discovery and search, event details and availability, participant registration, duplicate and capacity controls, participant/organizer/administrator functions, reporting, audit logging, validation, error handling, performance, reliability, security, and responsive/accessibility behavior.

Out of scope: Physical venue management, external payment settlement, institution-wide identity infrastructure, and third-party services not included in the approved baseline are excluded unless later integrated and documented.

Definitions: EMS - Event Management System; Participant - user who views/registers for events; Organizer - authorized user who creates/manages events; Administrator - authorized system-level user; SRS - Software Requirements Specification; RTM - Requirements Traceability Matrix; UAT - User Acceptance Testing.

2. Test Items

- Authentication and User module
- Event Management module
- Event Discovery UI
- Registration module
- Participant module
- Organizer module
- Administration module
- Reporting module
- Audit/Logging module
- Validation module
- API/UI Error Handler
- Authentication middleware and authorization controls
- Deployment/API communication layer
- Responsive React UI

3. Features to be Tested

The features below are taken from the SRS and are linked to their corresponding test cases:

SRS Req ID	Feature	Test Case
EMS-F-001	User registration	TC-REG-01
EMS-F-002	User authentication and role identification	TC-AUTH-01
EMS-F-003	Create event	TC-EVT-01
EMS-F-004	Edit authorized event	TC-EVT-02
EMS-F-005	Publish/cancel event	TC-EVT-03
EMS-F-006	Event listing	TC-VIEW-01
EMS-F-007	Search/filter events	TC-SEARCH-01
EMS-F-008	Event details and availability	TC-VIEW-02
EMS-F-009	Participant registration	TC-REG-02
EMS-F-010	Prevent duplicate registration	TC-REG-03
EMS-F-011	Capacity control	TC-REG-04
EMS-F-012	View personal registrations	TC-REG-05
EMS-F-013	Organizer participant management	TC-ORG-01
EMS-F-014	Administrator management	TC-ADM-01
EMS-F-015	Basic reports	TC-REP-01
EMS-F-016	Audit logging	TC-AUD-01
EMS-F-017	Input validation	TC-VAL-01
EMS-F-018	Error handling	TC-ERR-01
EMS-NF-001	API response performance	TC-PERF-01
EMS-NF-002	Availability/reliability	TC-OPS-01
EMS-NF-003	Server-side authorization	TC-SEC-01
EMS-NF-004	Timestamped audit/security logs	TC-LOG-01
EMS-NF-005	Responsive/accessibility UI	TC-UX-01
EMS-SR-001	Authenticated protected access	TC-SEC-01
EMS-SR-002	Role and ownership checks	TC-SEC-02

EMS-SR-003	HTTPS/TLS	TC-SEC-03
EMS-SR-004	Input validation/sanitization	TC-SEC-04
EMS-SR-005	Password hashing	TC-SEC-05

4. Features Not to be Tested

- Physical venue management, because it is outside the defined software baseline.
- External payment settlement, because payment integration is not part of the baseline scope.
- Institution-wide identity infrastructure and third-party identity services unless separately integrated and documented.
- Third-party SMS or other external services unless included through an approved change.
- Internal behavior of third-party libraries/frameworks; testing focuses on EMS usage and integration of those components.

5. Test Approach / Strategy

5.1 Test Levels

- Unit Testing - verify individual modules, validation functions, authorization logic, and business rules.
- Integration Testing - verify communication between React UI, Node.js APIs, Django components, database, authentication middleware, and logging components.
- System Testing - verify complete end-to-end EMS workflows from account access through event discovery and registration.
- Acceptance Testing (UAT) - verify that high-priority functional/security requirements and defined acceptance criteria are satisfied.

5.2 Test Types

- Functional testing - verify all functional requirements EMS-F-001 to EMS-F-018.
- Regression testing - re-run affected tests after defect fixes or changes.
- Performance testing - verify EMS-NF-001 under representative test load.
- Reliability/operational testing - verify startup, recovery from common failures, and availability target.
- Security testing - authentication, authorization, ownership checks, HTTPS/TLS, input sanitization, and password storage.
- Usability/accessibility testing - responsive layouts, labels, feedback, navigation, and keyboard interaction.
- Auditability testing - verify timestamped actor-linked records for important actions.

5.3 Entry Criteria

- Testable build is available and deployable.
- Required database, API, frontend, and Django components are available.
- Test accounts for Participant, Organizer, and Administrator roles are available.
- Representative event and registration test data are prepared.
- Test environment and defect tracking process are ready.

5.4 Exit Criteria

- 100% of planned test cases are executed.
- All high-priority functional and security requirements have passing referenced tests.
- No critical defects remain in core workflows.
- Acceptance criteria are satisfied or formally waived.
- RTM shows traceability from requirements to test cases.

5.5 Security Validation

- Verify authenticated access is required for protected participant, organizer, and administrator actions.
- Verify server-side role and ownership checks prevent unauthorized access or modification.
- Verify deployed client-server communication uses HTTPS/TLS.
- Use invalid and malicious input samples to verify validation and sanitization.
- Inspect credential storage to confirm passwords are not stored in plaintext and are protected using framework-supported password hashing.
- Verify security-relevant actions generate timestamped records associated with the acting user.

6. Test Environment

Hardware: Desktop/laptop or mobile device capable of running a modern web browser and network connection.

Software: React.js frontend, Node.js application/API services, Python/Django components, persistent database, modern web browser, and deployed/test API environment.

Tools: Postman or equivalent for API testing; browser developer tools; pytest/JUnit or applicable unit-test framework; JMeter or equivalent for performance testing; GitHub/Jira or similar for defect and test tracking; static/security analysis tools as configured by the project.

Test data: Participant, Organizer, and Administrator accounts; valid/invalid registration data; events with open/full/closed states; events with varying capacities; duplicate registration attempts; invalid dates/times/capacities; authorization boundary cases.

7. Test Schedule

Milestone	Planned Date
Test plan and test-case design	16-Sep-2026 to 18-Sep-2026
Test environment setup	18-Sep-2026 to 20-Sep-2026
Unit and integration testing	20-Sep-2026 to 23-Sep-2026
System and security test execution	23-Sep-2026 to 26-Sep-2026
Regression and defect verification	26-Sep-2026 to 27-Sep-2026
UAT / acceptance readiness	27-Sep-2026 to 28-Sep-2026

8. Test Deliverables

- Software Test Plan (this document)
- Functional and non-functional test cases
- Security validation test cases
- Test data and test scripts
- Test execution logs/results
- Defect reports and retest evidence
- Updated Requirements Traceability Matrix
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA / Test Lead	Ashmita Chaki	Prepare test plan, coordinate execution, monitor coverage and results.
Test Engineer	Aryan Mittal	Design and execute test cases, record evidence, log defects and perform retesting.
Test Engineer	Ashwath Rohan	Design and execute test cases, record evidence, log defects and perform retesting.

Developers	Atharv Kulkarni	Support test environment, analyze defects, fix defects and provide builds.
Product Owner / Evaluator	Course Coordinator	Review acceptance criteria and approve readiness/sign-off.

10. Risks and Mitigation

Risk	Mitigation
Delay in stable build delivery	Request early smoke builds and test modules incrementally.
Test environment/API downtime	Maintain a repeatable local/test environment and backup test data.
Unstable integration between frontend and APIs	Run integration tests early and use mocked/stubbed interfaces where appropriate.
Insufficient test data for capacity/duplicate scenarios	Prepare controlled event and participant datasets before execution.
Security defects discovered late	Execute security validation alongside functional testing rather than only at the end.
Regression after defect fixes	Run impacted test cases plus regression suite after each significant fix.

11. Assumptions & Dependencies

- The approved EMS SRS v1.0 remains the baseline unless a documented change is approved.
- React.js, Node.js, Python/Django, and database components are available in the test environment.
- Role-specific test accounts and representative event/registration data will be available before system testing.
- HTTPS/TLS can be enabled and inspected in the deployed test environment.
- Test tracking and source-control facilities used by the project remain available.

12. Suspension & Resumption Criteria

Suspend testing if:

- The test environment or critical API is unavailable and blocks execution of a substantial portion of the suite.
- The build is unstable enough that core workflows cannot be executed reliably.
- A blocking defect prevents execution of more than 30% of the planned test cases.
- Required test data or dependencies are unavailable.

Resume testing if:

- Blocking defects are resolved or an approved workaround is available.
- The environment and dependent services are stabilized.
- Required test data and accounts are restored.

13. Test Case Management & Traceability

The test cases use the same SRS requirement IDs and test IDs used in the project. Status is initially recorded as N (Not Executed); it should be updated to P (Passed) or A (Failed/Action required) during execution.

Test ID	SRS Req.	Type	Priority	Objective	Preconditions	Steps	Expected Result
TC-REG-01	EMS-F-001	Functional	High	Verify account registration with valid and invalid/missing fields.	Registration page available; test user not already registered.	1. Open registration form. 2. Submit valid required data. 3. Repeat with missing/invalid required fields.	Valid data creates an account; invalid/missing data is rejected with validation feedback.
TC-AUTH-01	EMS-F-002 / EMS-NF-003 / EMS-SR-001	Functional/Security	High	Verify login and correct role identification.	Participant, Organizer and Administrator test accounts exist.	1. Login with valid credentials for each role. 2. Attempt protected action without authentication.	Valid login creates authenticated session with correct role; unauthenticated protected request is rejected.

TC-EVT-01	EMS-F-003	Functional	High	Verify organizer can create an event with required details.	Authorized organizer account available.	1. Login as organizer. 2. Enter title, description, date, time, venue and capacity. 3. Save event.	Valid event is saved and appears in organizer event list.
TC-EVT-02	EMS-F-004	Functional	High	Verify authorized organizer can edit an owned event.	Owned event exists.	1. Open owned event. 2. Modify details. 3. Save. 4. View event as participant.	Updated information is persisted and displayed to users.
TC-EVT-03	EMS-F-005	Functional	High	Verify authorized publish/cancel actions.	Organizer/admin account and draft event available.	1. Publish event. 2. Verify listing. 3. Cancel event. 4. Verify listing/status.	Status changes persist and are reflected in event listings.
TC-VIEW-01	EMS-F-006	Functional	High	Verify event listing shows required summary information.	At least two events exist.	1. Open event listing. 2. Inspect event cards/list rows.	Users can view event titles, dates, venues and current status.
TC-SEARCH-01	EMS-F-007	Functional	Medium	Verify event search/filter returns matching results and handles no-result cases.	Events with distinguishable titles/dates/categories exist.	1. Search by title. 2. Filter by date/category. 3. Enter unmatched criteria.	Matching events are returned; unmatched criteria show a clear no-results state.
TC-VIEW-02	EMS-F-008	Functional	High	Verify event details and registration availability.	Event exists in open/full/closed states.	1. Open event details. 2. Inspect details and availability for each state.	Required details are shown and registration state is open/full/closed as applicable.
TC-REG-02	EMS-F-009	Functional	High	Verify authenticated participant can register for an open event with capacity.	Participant logged in; event open and not full.	1. Open event. 2. Select register. 3. Confirm submission. 4. Check registration record.	One participant-event registration record is created.
TC-REG-03	EMS-F-010	Functional	High	Verify duplicate registration is prevented.	Participant already registered for target event.	1. Attempt to register again for the same event.	Second attempt is rejected and no duplicate record is created.
TC-REG-04	EMS-F-011	Functional	High	Verify registration is blocked at configured capacity.	Event capacity reached; available capacity is zero.	1. Login as another participant. 2. Attempt registration.	Registration is blocked when capacity is zero.
TC-REG-05	EMS-F-012	Functional	Medium	Verify participant can view active and past registrations.	Participant has active/past registration records.	1. Open personal registrations. 2. Inspect records and statuses.	User can view registration records with event status.
TC-ORG-01	EMS-F-013	Functional/Security	High	Verify organizer can manage participants only for owned events.	Organizer owns Event A; Event B belongs to another organizer.	1. View Event A participant list. 2. Attempt to access Event B participant list.	Owned-event participant records are accessible; unauthorized event records are denied.
TC-ADM-01	EMS-F-014	Functional/Security	High	Verify administrator-only user/event management.	Administrator and non-admin accounts available.	1. Access admin functions as admin. 2. Attempt same functions as participant/organizer.	Admin actions are available only to authorized administrators.
TC-REP-01	EMS-F-015	Functional	Medium	Verify basic event and registration reports.	Events and registration records exist.	1. Open reporting. 2. Select authorized scope. 3. Generate report.	Report returns correct counts/details for selected scope.
TC-AUD-01	EMS-F-016 / EMS-NF-004	Functional/Audit	Medium	Verify important event and registration changes are logged.	Audit logging enabled.	1. Perform defined auditable action. 2. Inspect audit log.	Timestamped audit entry identifies the acting user and action.
TC-VAL-01	EMS-F-017 / EMS-SR-004	Functional/Security	High	Verify event date/time and capacity validation.	Organizer account available.	1. Enter past/invalid date or time. 2. Enter invalid capacity. 3. Submit.	Invalid schedule/capacity values are rejected with clear validation feedback.

TC-ERR-01	EMS-F-018	Functional	Medium	Verify clear errors for invalid input, insufficient permission and server failure.	Test environment can simulate each failure.	1. Submit invalid input. 2. Attempt unauthorized operation. 3. Trigger/handle server error.	Understandable error message is shown and no partial/duplicate record is created.
TC-PERF-01	EMS-NF-001	Performance	High	Verify 90th-percentile user-facing API response time is <= 3 seconds under project test load.	Production-like test environment and representative load available.	1. Run representative API load. 2. Collect response-time metrics. 3. Calculate 90th percentile.	90th percentile response time is <= 3 seconds.
TC-OPS-01	EMS-NF-002	Reliability	High	Verify deployment availability target and recovery from common failures.	Monitoring/logging available; deployment environment ready.	1. Exercise service startup/recovery. 2. Review availability measurement for project deployment.	Availability target is at least 99.0% excluding planned maintenance; common failures recover as expected.
TC-SEC-01	EMS-NF-003 / EMS-SR-001	Security	High	Verify protected API operations reject unauthenticated requests.	Protected API endpoint known.	1. Call protected endpoint without session/token. 2. Repeat with valid authenticated request.	Unauthenticated request is rejected; authenticated request is processed according to role.
TC-SEC-02	EMS-SR-002	Security	High	Verify server-side role and ownership enforcement.	Participant, two organizers and administrator available.	1. Attempt cross-role protected action. 2. Attempt organizer access to another organizer's event. 3. Attempt unauthorized modification.	User cannot access or modify records outside authorization scope.
TC-SEC-03	EMS-SR-003	Security	High	Verify deployed client-server communication uses HTTPS/TLS.	Deployed test environment available.	1. Access application/API. 2. Inspect connection/security details and network traffic.	Client-server communication is encrypted using HTTPS/TLS.
TC-SEC-04	EMS-SR-004	Security	High	Verify malicious/invalid inputs are safely rejected or handled.	Validation endpoints/forms available.	1. Submit malformed and malicious-looking input values in relevant fields. 2. Observe processing and stored data.	Inputs are rejected or safely handled without unintended execution.
TC-SEC-05	EMS-SR-005	Security	High	Verify passwords are not stored in plaintext.	Test database access available to authorized test team.	1. Create/change a test password. 2. Inspect credential storage through authorized test access.	Stored credential is a password hash rather than plaintext.
TC-LOG-01	EMS-NF-004	Auditability	Medium	Verify authentication, event, registration and administrative actions create timestamped logs.	Logging enabled; test accounts available.	1. Perform representative actions. 2. Inspect logs.	Required events appear with timestamp and actor information.
TC-UX-01	EMS-NF-005	Usability/Accessibility	Medium	Verify responsive layout and basic accessibility practices.	Supported desktop/mobile viewport sizes available.	1. Inspect UI at desktop and mobile sizes. 2. Check labels, feedback, navigation and keyboard controls.	UI remains understandable; labels/feedback are meaningful and applicable controls are keyboard accessible.

14. Requirements Traceability Matrix (RTM)

The RTM below shows the link between each SRS requirement and its test case. The initial status is N (Not Executed).

Req ID	Requirement Short	Test Case(s)	Status
EMS-F-001	User registration	TC-REG-01	N

EMS-F-002	User authentication	TC-AUTH-01	N
EMS-F-003	Create event	TC-EVT-01	N
EMS-F-004	Edit event	TC-EVT-02	N
EMS-F-005	Publish/cancel event	TC-EVT-03	N
EMS-F-006	Event listing	TC-VIEW-01	N
EMS-F-007	Search/filter events	TC-SEARCH-01	N
EMS-F-008	Event details/availability	TC-VIEW-02	N
EMS-F-009	Participant registration	TC-REG-02	N
EMS-F-010	Prevent duplicate registration	TC-REG-03	N
EMS-F-011	Capacity control	TC-REG-04	N
EMS-F-012	View personal registrations	TC-REG-05	N
EMS-F-013	Organizer participant management	TC-ORG-01	N
EMS-F-014	Administrator management	TC-ADM-01	N
EMS-F-015	Basic reports	TC-REP-01	N
EMS-F-016	Audit logging	TC-AUD-01	N
EMS-F-017	Input validation	TC-VAL-01	N
EMS-F-018	Error handling	TC-ERR-01	N
EMS-NF-001	API response target	TC-PERF-01	N
EMS-NF-002	Availability target	TC-OPS-01	N
EMS-NF-003	Authorization enforcement	TC-SEC-01	N
EMS-NF-004	Timestamped audit/security logs	TC-LOG-01	N
EMS-NF-005	Responsive/accessibility	TC-UX-01	N
EMS-SR-001	Authenticated protected access	TC-SEC-01	N
EMS-SR-002	Role and ownership checks	TC-SEC-02	N
EMS-SR-003	HTTPS/TLS	TC-SEC-03	N
EMS-SR-004	Input validation/sanitization	TC-SEC-04	N
EMS-SR-005	Password hashing	TC-SEC-05	N

15. Test Metrics & Reporting

The main metrics we will collect are:

- Percentage of planned test cases executed.
- Percentage of test cases passed/failed.
- Requirement coverage across functional, non-functional and security requirements.
- Defect count and defect density.
- Defect aging and open/closed defect counts.
- Performance response-time measurements, including 90th percentile.
- Availability/recovery measurements.
- Security test coverage and unresolved security defects.

The reports we will prepare are:

- Daily/iteration test execution status
- Defect and retest reports
- Final Test Summary Report

16. Final Acceptance Criteria

- All planned test cases are executed and results recorded.
- All high-priority functional and security requirements have passing tests or formally approved exceptions.
- EMS-NF-001 performance target is met under the defined project test load.
- EMS-NF-002 availability target is addressed with evidence for the project deployment environment.
- EMS-NF-005 responsive/accessibility checks pass for supported screen sizes and applicable keyboard interactions.
- No critical defects remain in core workflows at acceptance readiness.
- RTM demonstrates complete traceability from SRS requirements to test cases.

17. Approvals

Role	Name	Signature / Date
QA / Test Lead	Ashmita Chaki	Ashmita Chaki 16-09-2026
Test Engineer	Aryan Mittal	Aryan Mittal 16-09-2026
Test Engineer	Ashwath Rohan	Ashwath Rohan 16-09-2026
Development Lead	Atharv Kulkarni	Atharv Kulkarni 16-09-2026
Product Owner / Course Coordinator		